



Combined Reuse of Paint Industry and Cattle Slaughterhouse Effluents for the Fertigation of Mombaça Grass

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Abstract This study aimed to evaluate the effects of the combined application of effluents from the paint industry and cattle slaughterhouses on the fertigation of Mombaça grass (*Panicum maximum* cv. Mombaça). The experiment was conducted in pots with 12 plants using a completely randomized design in a 5×2 factorial arrangement, with five sampling times (0, 2, 4, 6, and 8 months) and evaluated under two conditions: with and without application of the combined effluent at a 1:1 ratio. Fertigation promoted a significant increase in Mombaça grass productivity, reaching an average of 18.28 t/ha of dry matter. Greater extraction of nitrogen, calcium, phosphorus, potassium, and magnesium was observed over 12 months, even with continuous plant extraction. An increase in potential cation exchange capacity was also observed, indicating improved soil fertility. In contrast, there was a decreasing trend in pH. The effective cation exchange capacity (ECEc) remained

stable, revealing a dynamic balance between nutrient input and absorption. The data demonstrate the agronomic potential of reusing these effluents as a sustainable alternative for fertigation yet highlight the importance of long-term studies to assess potential environmental risks and contaminant accumulation. This study contributes to the integration of liquid-waste management and sustainable agricultural practices.

Keywords Fertigation · Industrial effluent · Soil fertility · *Panicum maximum*

1 Introduction

The intensification of agriculture in the face of the growing demand for food, fiber, and energy has imposed significant challenges on the sustainability of production systems, especially in the face of water scarcity and the increasing generation of industrial waste (FAO, 2022). Notably, the agricultural reuse of effluents emerges as a strategic alternative, associating the use of wastewater with the supply of nutrients to the soil, promoting the conservation of natural resources and complementary fertilization (Kiziloğlu et al., 2020; Santos et al., 2023).

Paint manufacturing and cattle slaughtering industries consume large quantities of water and, consequently, generate significant volumes of effluent (Martins et al., 2025; Nair et al., 2021), making them

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subjects of great interest for agricultural reuse. However, these effluents present concerning environmental limitations: paint industry effluent contains toxic metals and compounds derived from organic solvents (Nair et al., 2021), while cattle slaughterhouse effluent has high concentrations of sodium and pathogenic microorganisms (Freitas et al., 2026). Although fertigation is an economically viable reuse method, its application to the soil requires caution, particularly due to potential risks associated with nutrient leaching, salinization, contaminant mobilization, impacts on groundwater quality, and surface runoff into water bodies.

Effluent from the cattle slaughter industry is widely used in fertigation, especially in pastures, because of its high content of organic matter, nitrogen and phosphorus (Pereira AKS et al., 2024), in addition to other macro- and micronutrients that can significantly influence the chemical attributes of the soil, such as pH, potential acidity ($H^+ + Al^{3+}$), cation exchange capacity (CEC), and base saturation, thereby impacting fertility, nutrient availability and soil's ionic balance (Lima et al., 2023; Nascimento et al., 2022; Yang et al., 2021).

Effluents from the paint industry are rarely used in fertigation systems because they are highly toxic. However, ecotoxicological studies on planarians (*Girardia tigrina*) have shown that raw effluent from the paint industry was not toxic to this organism at a contact time of 96 h (Carlos et al., 2025), whereas a solution containing 26.8% cattle slaughter effluent resulted in the death of 50% (LC_{50}) of planarians at the same exposure time. It is also worth noting that the evaluated slaughter effluent had been previously treated through screening, fat separation boxes, and facultative lagoons (Pereira MAB et al., 2024). Costa et al. (2024) evaluated the ecotoxicity of cattle slaughter and paint industry effluents combined (1:1) and reported that the LC_{50} occurred with a solution containing 60.76% of the mixture; in other words, the addition of paint effluent resulted in a reduction in the toxicity of the final mixture compared with the isolated use of cattle slaughter effluent.

From this perspective, this study evaluated the effects of the combined application of these effluents on forage cultivation systems, as the paint industry effluent is composed of macro- and micronutrients that can be used for plant production, mainly carbonates (Carlos et al., 2024). Furthermore,

Goldstein-Plesser et al. (2026) describe grasses as effective bioindicators of contamination, as they exhibit various stress responses when exposed to environmental stressors such as heavy metals. The primary site of plant-metal interaction is the root apical meristem, where the metal inhibits growth and disrupts typical root formation (Angulo-Bejarano et al., 2021). Thus, the development of grasses can also serve as a bioindicator of soil toxicity.

Mombaça grass (*Panicum maximum* cv. Mombaça) was used in the present study. This species exhibits high productivity, is sensitive to variations in soil fertility, and has a high nutrient extraction capacity, making it an efficient bioindicator plant (Mesquita et al., 2022). This grass responds significantly to the availability of macro- and micronutrients, and is influenced by conditions such as soil acidity and salinity, which are frequently altered by the application of effluents (Melo et al., 2021).

Thus, understanding how soil chemical attributes change as a result of the application of combined effluents from the paint and cattle slaughter industries is essential to validate reuse as a sustainable and safe practice. Furthermore, reports on studies involving the combined application of effluents are scarce, and there is a need for information grounded in scientific rigor to encourage this reuse alternative, aiming for contaminant dilution and nutrient complementarity. This study aimed to evaluate the impact of this application on the fertility of soil cultivated with mombaça grass, with an emphasis on the nutrient content, acidity, pH, and cation exchange capacity, as well as possible effects of the combined effluent on plant growth.

2 Materials and Methods

2.1 Soil Used for the Study

The experiment was conducted in a greenhouse at the Instituto Tocantinense Presidente Antonio Carlos – ITPAC Porto Nacional Faculdade Presidente Antônio Carlos (ITPAC), located in Porto Nacional—TO, at the geographic coordinates 10° 69' 57" S, 48° 8' 48" W, at an altitude of 212 m. The soil, classified as Red-Yellow Latosol, was stored in 5 L pots, and its physical and chemical characteristics are presented in Table 1. No soil correction was carried out, since Mombasa grass grows well naturally in Red-Yellow

Table 1 Chemical characteristics of the soil in the 0 to 20 cm layer in Red-Yellow Latosol. Porto Nacional, TO, Brazil

pH (H ₂ O)	P mg/dm ³	K	Ca cmolc/dm ³	Mg	Al	H + Al	V (%)	O.M g/dm ³
5.8	45.3	0.05	0.03	0.22	0.21	3.05	9.16	13.5

O.M. = organic matter; Clay (HMFS + Na g/kg) = 421.6; silt g/kg estimated = 91.2; total sand (sieve g/kg) = 487.2 (particle-size analysis by the pipette method); pH in H₂O, KCl, CaCl₂—ratio 1:2.5; P, Na, K, Fe, Zn, Mn, Cu—Mehlich extractant 1.81; Ca, Mg, Al—KCl extractant 1 mol/L; H + Al—calcium acetate extractant 0.5 mol/L, pH 4.20

Latosol, the predominant soil type in the Brazilian Cerrado biome.

2.2 Effluent From the Paint and Cattle Slaughter Industry Used for Fertigation

Effluents were collected from regional industries, with the cattle slaughter effluent coming from a slaughterhouse with an average slaughter of 1,200 heads of cattle per month and a small paint industry that manufactures water-based paints, oil-based paints, and varnishes. In this paint industry, lead- and chromium-based inputs are not used in the production process, as the paints are intended for use in civil construction. Only the cattle slaughter effluent underwent prior treatment, consisting of a grating, a fat separation box, and three optional lagoons. After collection, the samples were characterized using the parameters listed in Table 2. For elemental analysis, a 10 ml aliquot of the samples was separated, acidified with 5% suprapur nitric acid, and then filtered through a 0.45 µm membrane. After reading the multi-element standard curve, aliquots (acidified and filtered) were analyzed using Plasma Atomic Emission (ICP-AES) equipment, with a limit of quantitation (LOQ) of 0.01 mg/L.

2.3 Preparation of Mombaça grass samples

2.3.1 Grass Cultivation

Mombaça grass was selected as the primary plant for this study because of its greater need for nutrients, demand for deep soil, good drainage systems, and adaptation to the local climate, which is characterized by high temperatures and > 1,000 mm of rainfall annually. Mombaça grass was planted with approximately 30 seeds per pot. After 18 d of sowing, thinning was performed, leaving 12 plants in each pot. The pots were organized in rows with a spacing of

1 m between them. The distance between the pots within each row was 0.5 m.

The experiment was implemented in a completely randomized experimental design in a 5×2 double factorial scheme, consisting of different collection times (T1=0 months, T2=3 months, T3=6 months, T4=9 months, and T5=12 months) with and without the application of the combined effluent (1:1), with five replicates.

The crop water requirements in all pots were calculated using the daily reference evapotranspiration. The crop water demand in all plots was calculated using daily reference evapotranspiration (ET_0) and the Penman–Monteith method: $ET_0 \times K_c = ET_c$ (Allen, 2006). Crop evapotranspiration (ET_c) was calculated using the daily values of the crop coefficient (K_c). Accumulated water depths were applied daily to replenish ET_c .

2.3.2 Fertigation

Once the combined effluent water depths were established for application to the experimental plots, 60 kg/ha of nitrogen was used as the reference concentration to calculate these depths. Water depth and fertigation were determined by weighing the pots, and the difference in weights corresponded to the volume applied to increase the field capacity. The combined effluent was applied using manual watering to promote contact with grass leaves, simulating application through sprinkler systems. The first dose of the combined effluent was applied 20 d after planting the grass, and subsequent applications were performed once a week.

2.3.3 Productivity of Mombaça grass

Grass cuttings were performed at 30, 90, 180, 270, and 360 days after transplanting, harvesting was consistently carried out every 30 days—coinciding

Table 2 Physical–chemical and microbiological characterization of effluents

Parameters	Combined effluent (1:1)	Method*	Reference ¹ ISO/ ² Ayers&Westcot/ ³ ANZG
pH	7.33 ± 0.05	Potentiometric	² 6.5–8.4
COD (mg/L)	699 ± 40	Spectrophotometric	
BOD (mg/L)	74 ± 12	Respirometric	¹ ≤ 20 (Average)–35 (Máx)
DO (mg/L)	5.28 ± 0.5	Oximetric	
Conductivity (µS/cm)	533.4 ± 3.5	Conductometric	² 700
Total dissolved solids (mg/L)	240.5 ± 7.8	Gravimetric	² < 450
Total suspended solids (mg/L)	122.5 ± 19	Gravimetric	¹ ≤ 30 (Average)–50 (Máx)/ ² < 50
Total solids (mg/L)	363 ± 11	Gravimetric	
Turbidity (NTU)	72 ± 5	Nephelometric	
Total nitrogen (mg/L)	1262 ± 63	Spectrophotometric	
Sodium (mg/L)	28.04 ± 0.63	Atomic emission	² SAR < 3.0
Potassium (mg/L)	14.24 ± 0.03	Atomic emission	
Magnesium (mg/L)	4.93 ± 0.1	Atomic emission	
Calcium (mg/L)	3.85 ± 0.33	Atomic emission	
SAR	2.26	Calculated	² 0–15
Molybdenum (mg/L)	< 0.01	Atomic emission	^{2,3} 0.01
Phosphorus (mg/L)	10.33 ± 0.31	Atomic emission	
Zinc (mg/L)	0.12 ± 0.00	Atomic emission	^{2,3} 2.0
Iron (mg/L)	0.12 ± 0.00	Atomic emission	² 5.0/ ³ 0.2
Copper (mg/L)	0.2 ± 0.00	Atomic emission	^{2,3} 0.20
Nickel (mg/L)	< 0.01	Atomic emission	^{2,3} 0.2
Manganese (mg/L)	0.1 ± 0.00	Atomic emission	^{2,3} 0.2
Cobalt (mg/L)	< 0.01	Atomic emission	^{2,3} 0.05
Lithium (mg/L)	< 0.01	Atomic emission	² 2.5/ ³ 0.075
Total coliforms (CFU/100 mL)	8000	Atomic emission	¹ ≤ 10,000
<i>E. coli</i> (CFU/100 mL)	900	Filter membranes	¹ ≤ 10,000/ ³ 100

APHA (2023). SAR = Sodium Adsorption Ratio— $SAR = \frac{[Na^+]}{\sqrt{\frac{[Ca^{2+}] + [Mg^{2+}]}{2}}}$, using values in mmol_c

¹ISO 16075–2/2020

²Ayers&Westcot (1985)

³ANZG: Water Quality for Irrigation and General Water Uses: Guidelines. Australian and New Zealand Guidelines for Fresh and Marine Water Quality (2024)

with the onset of the flowering phase—at a height of 45 cm above the soil, across the five replicates of each treatment. After collection, the plant material was dried in an oven with forced air circulation at 65 °C for a period of 72 h. The samples were then ground in a Wiley-type mill to determine the dry matter yield and for subsequent analysis of the macronutrient contents (nitrogen [N], phosphorus [P], potassium [K], calcium [Ca], and magnesium [Mg]) based on the methodology described by Silva (2009).

2.3.4 Assessment of Soil Chemical Properties

Possible changes in soil chemical attributes were investigated by collecting five soil samples up to a depth of 20 cm in each experimental plot at two-month intervals. After collection, the soil samples were sent to the laboratory, and following the methodology proposed by Silva (2009), the pH, phosphorus, potassium, calcium, magnesium, potential acidity, effective and potential cation exchange capacities, and base saturation were determined. As the soil analysis was destructive, the volume of soil collected

was 100 g per pot to minimize the influence of soil removal.

The data on soil chemical attributes obtained in each experimental unit were statistically analyzed following the methodology presented by Ferreira (2018), in which the data were subjected to analysis of variance, regression, and mean tests. The means were compared using Tukey's test for comparisons between samples and Dunnett's test for comparing treatments with the control sample, both at a 5% probability level. The statistical program SISVAR® (Ferreira, 2018) was used to perform the statistical analyses.

3 Results and Discussion

Regarding the effluent characteristics, it is observed that the electrical conductivity (EC) and sodium adsorption ratio (SAR) meet the irrigation requirements described by Ayers and Westcot (1985). However, regarding soil salinization, the infiltration rate, which relates both EC and SAR, should also be considered. In this case, the degree of restriction on effluent use for irrigation was classified as mild to moderate. Salinity increases the infiltration rate; however, a high SAR value (sodium concentration relative to calcium and magnesium concentration) decreases the infiltration rate. Consequently, a low infiltration rate promotes the proliferation of vectors (mosquitoes), while a high infiltration rate allows for greater leaching, which may lead to groundwater contamination or soil accumulation. With high salinity and high SAR, both problems can occur simultaneously (Ayers & Westcot, 1985). Based on the SAR calculation, it is also possible to establish that the use of effluent in irrigation does not promote soil toxicity due to sodium.

Pereira MAB et al., (2024) conducted studies with bovine slaughter effluent, and according to the recorded concentrations of sodium, calcium, and magnesium, the SAR of the effluent was 4.38 and the EC was 914 $\mu\text{S}/\text{cm}$. Based on data from Estime et al., (2026) and Freitas et al., (2026), the SAR values obtained for bovine slaughter effluent were 7.2 and 11.6, and EC values were 1503 and 1884 $\mu\text{S}/\text{cm}$. Consequently, the combination of cattle slaughter effluent with paint industry effluent (1:1) resulted in a RAS value < 3.0 and EC $< 700 \mu\text{S}/\text{cm}$, that is, below

the maximum limit recommended for irrigation water (Ayers & Westcot, 1985).

In addition, the combination of effluents promoted the reduction of *E. coli*, meeting the value recommended by ISO 16075–2/2020, but was not sufficient to reach the level required by ANZG (2024). In the study by Estime et al., (2026) the presence of this microorganism in cattle slaughter effluent ranged from 2.10^4 – 3.10^5 CFU/100 mL, therefore, taking these values as a reference, the combination of effluents represented a decay of 1.3 to 2.5 logs.

Quantification of BOD demonstrated that even after combining the effluents, the biodegradable organic matter content was high, according to the limits recommended by ISO 16075–2/2020; however, a reduction in the organic load was observed after mixing. Freitas et al. (2026) recorded BOD values of 142 mg/L for cattle slaughter effluent; consequently, the combination with paint effluent promoted a dilution of the organic load to 74 mg/L. The TSS (Total suspended solids) parameter also exceeded the recommended value (< 50 mg/L) for use in drip irrigation systems (Ayers & Westcot, 1985). Carlos et al. (2025) obtained a TSS value of 73 mg/L for paint industry effluent, while Estime et al. (2026) reported values between 57.5 and 123 mg/L for cattle slaughter effluent. BOD and TSS parameters can contribute to the clogging of irrigation systems through the formation of biofilms and deposition of particles. In addition, the transport of this effluent by surface runoff or percolation into the soil can pose risks to water bodies.

The oligonutrients evaluated in the physicochemical characterization did not exceed the values established by ISO 16025–2/2020 and ANZG (2024) for irrigation water.

3.1 Productivity of Mombaça Grass

The analysis of variance indicated a significant effect ($p < 0.01$) of the combined application of effluents from the paint industry and cattle slaughter via fertigation on forage productivity. The differences between treatments based on the mean comparison tests are presented in Table 3.

The data presented in Table 3 show that an increase exists in the productivity of Mombaça grass (*Panicum maximum* cv. Mombaça) in the treatments subjected to fertigation with effluent from the paint

Table 3 Estimated average productivity of Mombaca grass as a function of fertigation with effluent from the paint industry and cattle slaughter combined for agricultural reuse

Period (Months)	Average productivity (t/ha)*			
	Control	Combined effluent	Average $\pm$ SD	Average $\pm$ SD
1	10.04a	7.2c	7.24 $\pm$ 2.17	28.54
3	1.98b	10.45b	5.72 $\pm$ 3.87	70.07
6	3.60b	10.07c	6.88 $\pm$ 3.13	45.6
9	5.06b	11.73b	6.55 $\pm$ 2.66	47.23
12	5.04b	18.28a	8.74 $\pm$ 5.74	60.41

Averages followed by the same letter in the columns do not differ statistically from each other by the Tukey's test at 5% probability. CV = coefficient of variation

* The amounts per period are not cumulative

industry and cattle slaughter combined for 12 months compared to the control treatment (without effluent application). The average productivity of the fertigated treatments was 18.28 t/ha of dry matter, significantly outperforming the control group. Martins et al. (2025), using only cattle slaughter effluent, reached 15.20 t/ha. This productive performance reinforces the potential of the effluent as a source of essential nutrients for forage growth, particularly nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), and magnesium (Mg), which are continuously extracted by plants throughout cutting cycles.

Oliveira et al. (2019) reported that the application of slaughterhouse effluent rich in organic matter and nitrogen compounds significantly favors the accumulation of biomass in tropical grasses, especially in cultivars with high nutritional requirements, such as Mombaca grass. Similarly, Souza et al. (2020) observed that fertigation with effluents of animal origin and effluents rich in carbonates increased the dry mass production as well as the nutrient concentrations in plant tissues, evidencing greater efficiency in the absorption and use of the elements available in the soil.

The dynamics of nutrient extraction by mombaca grass over 12 months were directly associated with the vigorous growth and high regrowth rate characteristics of the species. In environments with a regular nutrient supply, such as fertigation treatments with effluent, plants have a greater capacity to absorb and redistribute essential elements, especially N and K, which are in high demand for protein synthesis, chlorophyll formation, and osmotic regulation (Santos et al., 2021). In addition, a continuous supply of

phosphorus contributes to the development of the root system, increasing the efficiency of water and nutrient capture, particularly in soil columns with a good physical structure.

In experiments conducted in Colombia and at the international level, Ramírez et al. (2020) reported that the regular application of effluents from cattle slaughterhouses in forage production systems can increase productivity by up to 120% compared with areas managed only with mineral fertilizers, provided that there is control over the organic load and soil salinity levels.

3.2 Nutrient Extraction By Mombaca Grass

The average removal of nitrogen (N), phosphorus (P), potassium (K), sodium (Na), calcium (Ca), and magnesium (Mg) were calculated based on the productivity and concentration of nutrients in the plant dry matter (Table 4).

In plants irrigated only with water (control), a general reduction in the productivity of Mombaca grass was observed (Table 3), which resulted in the absence of statistically significant differences in the removal of N, P, Ca, and Mg between collection months (Table 4). This behavior may be associated with the partial depletion of nutrients available in the soil, or with the physiological adaptation of the plant in the advanced stages of the cultivation cycle, which tends to prioritize the accumulation of structural biomass to the detriment of the intensive absorption of macronutrients. Studies by Oliveira et al. (2019) and Ramírez et al. (2020) also identified stabilization or a decrease

Table 4 Average removal of nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), and magnesium (Mg) by Mombaça grass resulting from fertigation with effluent from the paint industry and cattle slaughter combined for agricultural reuse

Period (Months)	N	P	K	Ca	Mg
Control (t/ha)					
1	35.72a	1.02a	3.42a	2.30a	0.83a
3	31.48a	0.52a	3.20a	2.14a	0.78a
6	30.73a	0.47a	2.78a	1.96a	0.72a
9	30.62a	0.41a	2.53a	1.82a	0.67a
12	29.79a	0.37a	1.83a	1.59a	0.56a
Average	31.67	0.56	2.75	1.96	0.71
CV (%)	25.43	48.05	11.24	19.52	8.6
Combined effluent (1:1) (t/ha)					
1	88.69b	2.25a	16.90a	23.43b	8.67b
3	248.98a	2.44a	10.65a	18.35b	7.79b
6	234.52a	5.25a	16.68a	17.18b	6.36b
9	176.66ab	5.13a	12.70a	21.17b	7.83b
12	216.55a	5.76a	15.93a	34.38a	12.72a
Average	193.56	3.73	14.57	31.38	8.47
CV (%)	18.56	52.12	30.26	16.01	30.1

Averages followed by the same letter in the columns do not differ statistically from each other by the Tukey's test at 5% probability

in nutrient extraction by tropical grasses under prolonged irrigation, especially when there was no adequate replacement of the nutrient solution or careful irrigation management.

However, the results of the mean comparison test showed statistically significant differences ($p < 0.05$) in the extraction of nitrogen (N), calcium (Ca), and magnesium (Mg) from Mombaça grass (*Panicum maximum* cv. Mombaça) among the different treatments (Table 4). The highest extraction values of these macronutrients were observed in the treatments with cattle slaughter effluent after 3 and 12 months of cultivation, which significantly exceeded the levels recorded in the control treatment (without effluent application).

The greater accumulation of N, Ca, and Mg over 12 months can be attributed to the continuous availability of these elements in the soil promoted by fertigation with effluent, which contained high levels of nitrogen compounds, carbonates, and minerals from organic waste, and metabolites from animal and industrial processing. Studies by Oliveira et al. (2019)

and Souza et al. (2020) indicate that cattle slaughterhouse effluent is a rich and efficient source of essential nutrients, favoring the growth of the aerial part and the efficiency in the absorption of divalent cations such as Ca^{2+} and Mg^{2+} , essential for the stability of cell membranes, enzymatic activity, and cell division in tropical grasses. Furthermore, the characteristics of the combined effluent indicated a slight to moderate restriction regarding changes in the infiltration rate (Ayers & Westcot, 1985), which may have contributed to the accumulation of these nutrients in the soil.

The marked nitrogen extraction, especially in the initial treatments (up to 3 months), is directly related to the role of this nutrient in the formation of structural proteins and in stimulating sprouting and tillering, which are characteristics that define the initial vigor of Mombaça grass. The greater accumulation of Ca and Mg throughout the 12-month cycle was associated with the continuous need for these elements to maintain cellular integrity, nutritional balance, and pH regulation in the root apoplast, as reported by Santos et al. (2021) in long-term experiments with grasses under different fertigation regimes.

In a study conducted by Ramírez et al. (2020), similar behavior was observed in the response of *Panicum maximum* subjected to the continuous application of cattle slaughterhouse effluent, with a significant increase in N, Ca, and Mg concentrations in plant tissues compared to conventional mineral fertilizers. These authors also highlighted that, in addition to nutritional supply, the rational use of effluent contributes to improving the physicochemical properties of the soil, favoring the efficiency of root absorption.

3.3 Chemical Properties of Soil

The factors corresponding to the use of the combined effluent, as well as the evaluation periods, resulted in changes in the chemical attributes of the soil; Fig. 1 illustrates the variations in pH and base saturation.

Figure 1, it can be observed that over the 12 months of cultivation under continuous application of combined effluents from the paint industry and cattle slaughter (1:1), a progressive reduction in the pH and base saturation (V%) values was noted in the evaluated soils, both in the control group and in the plots treated with the effluents. These results indicate

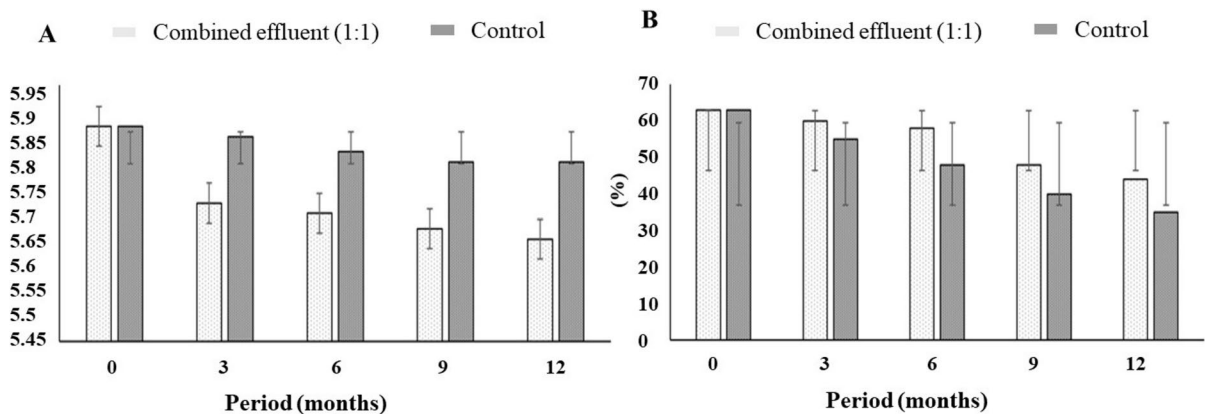


Fig. 1 **A** Levels of hydrogen potential (pH) and **(B)** base saturation (V%) of the soil, as a function of fertigation with effluent from the paint industry and cattle slaughter combined for agricultural reuse

a possible acidifying effect of this management practice, even though an increase in alkalinity might be expected owing to the presence of carbonate ions in the paint industry effluent (Carlos et al., 2024).

The decrease in soil pH over time may be associated with the mineralization of organic matter present in effluents, the oxidation of nitrogenous compounds such as ammonia, and the release of organic acids, phenols, and other industrial chemical residues, especially in the case of paint industry effluents. Based on Huang et al. (2023), the application of industrial effluents can promote soil acidification, especially when the material contains organic compounds with a high oxidizable charge, affecting the acid–base balance of the system.

From an agronomic perspective, the reduction in base saturation is concerning, as it implies a decrease in the availability of essential cations (Ca^{2+} , Mg^{2+} , K^+ , and Na^+) in the cation exchange capacity (CEC) of the soil. This can compromise fertility and negatively affect crop development. Silva et al. (2022a, b) also reported a V% reduction in soils treated with organic effluents of animal origin, revealing that base leaching and replacement by H^+ and Al^{3+} ions may occur with the intensification of applications in continuous systems.

This behavior may be related to the high nutritional requirements of forage, especially in intensive production systems. Mombaça grass is known for its high production potential, accumulating, per cycle, approximately 300 to 400 kg/ha of nitrogen, 40 to 50 kg/ha of phosphorus, and up to 450 kg/

ha of potassium, in addition to significant amounts of calcium and magnesium, estimated at 250 and 90 kg/ha, respectively, depending on management and soil and climate conditions (Euclides et al., 2021; Werner et al., 2022).

Given this high nutrient demand, the contribution of effluent application, especially in the first years of cultivation, may not be sufficient to significantly alter soil nutrient availability, particularly when the application rate is moderate. However, some elements tend to gradually accumulate with continued use, which can generate more significant effects in the medium-to long-term, especially when combined with appropriate fertilizer management and efficient nutrient cycling.

However, some authors have reported that the pH and base saturation values remain unchanged or even increase in soils treated with agro-industrial effluents, especially when the residues have high concentrations of carbonates or metal oxides. Rodrigues et al. (2021) observed an increase in the pH of sandy soils treated with meat-packed plant effluent, owing to the presence of calcium residues and the alkalinity of the effluent. This divergence highlights the importance of chemically characterizing effluents before their application and monitoring their long-term effects.

The acidifying behavior observed in this study reveals the need for corrective management measures, such as the periodic application of agricultural amendments (limestone) to neutralize the generated acidity and adjustments to the frequency and dose of

effluent application. Furthermore, continuous acidification can increase the mobility of heavy metals in the soil, which requires attention to ensure the environmental safety of agricultural reuse.

Figure 2 reports the average values for the evaluation of phosphorus (P), potassium (K), calcium (Ca), and magnesium (Mg) contents over 1 year of cultivation with Mombaça grass (*Panicum maximum* cv. Mombaça) under different management conditions, highlighting the marked differences between the treatments with and without the application of combined effluents.

Figures 2A–D show that while the control group maintained relatively constant values with small decreasing fluctuations over the months of evaluation, treatments with the combined application of effluents from the paint industry and cattle slaughter showed significant increases in the levels of these nutrients, with subsequent stabilization over time.

Soil available phosphorus levels increased considerably in the effluent treatments in the first few months and remained stable in subsequent months (Fig. 2A). In contrast, in the control group, phosphorus levels decreased slightly, reflecting the effects of

crop extraction and the lack of replenishment. These results reveal that effluent application is effective in providing bioavailable phosphorus to the soil, which is particularly relevant given that this nutrient has low mobility and high adsorption in tropical soils, especially in acidic soils, where it is more readily absorbed by iron and aluminum oxides (Moreira; Siqueira, 2021).

The stability of phosphorus levels over time in soils with effluent may also be related to the gradual mineralization of organic phosphorus present in cattle effluent, which, based on Lima et al. (2023), constitutes an important source of slow- and continuous-release phosphorus, helping meet plant demands even under intense vegetative growth. Notably, Mombaça grass has a high rate of phosphorus accumulation, especially in the initial stages of regrowth, which reinforces the importance of the continuous replenishment of this nutrient.

Potassium, an essential nutrient for osmotic regulation, cell growth, and enzyme activation, exhibited a similar behavior to phosphorus: stability in the control group, with a slight downward trend and a substantial increase, followed by stabilization in the

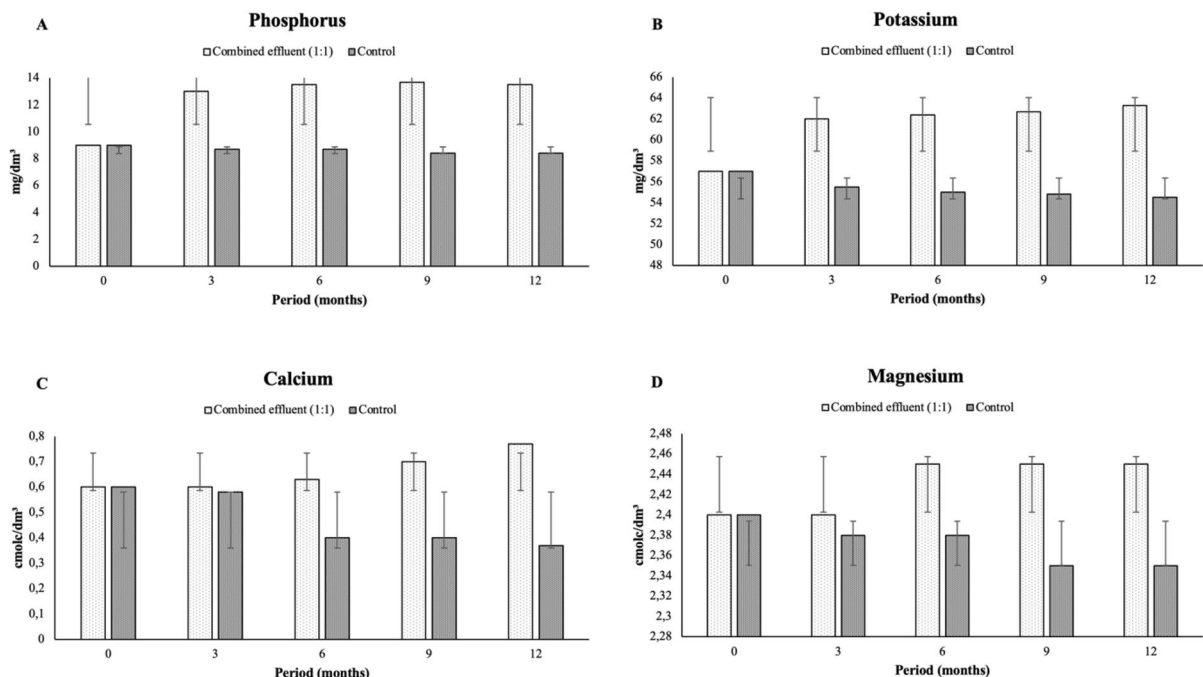


Fig. 2 Levels of phosphorus, potassium, calcium, and magnesium in the soil, as a function of fertigation with effluent from the paint industry and cattle slaughter combined for agricultural reuse

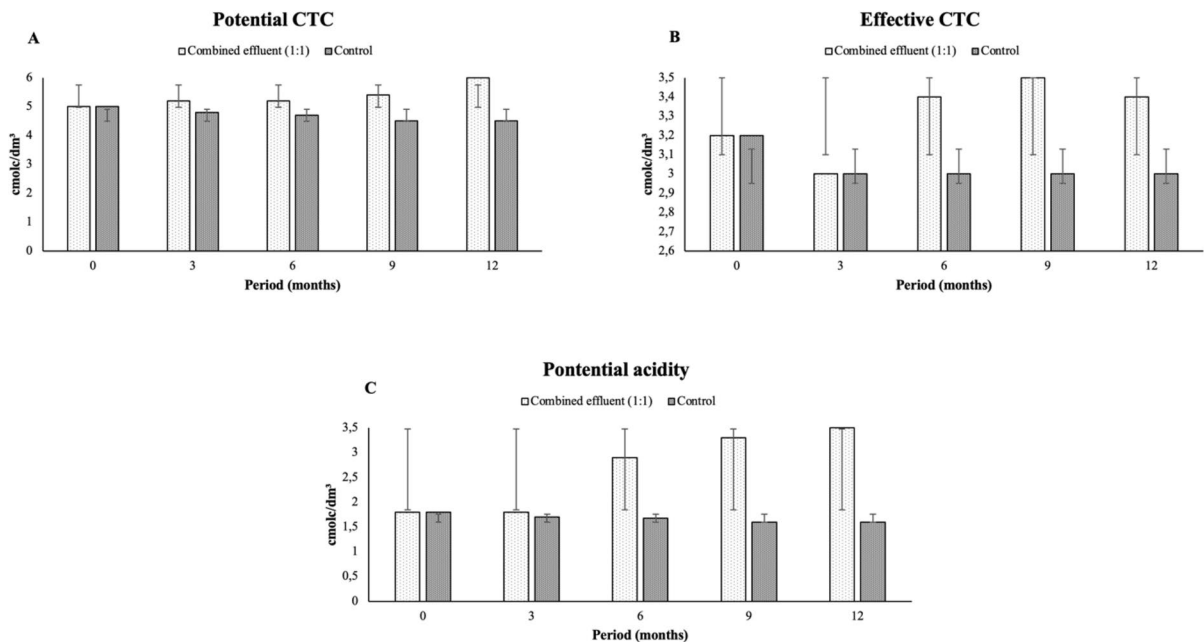


Fig. 3 Levels of CECp, CECe, and potential soil acidity as a function of fertigation with effluent from the paint industry and cattle slaughterhouse combined for agricultural reuse

effluent treatments (Fig. 2B). Because potassium is highly soluble and mobile in soil, its dynamics are closely related to leaching and plant uptake.

According to Silva et al. (2022a, b), slaughterhouse and meat-processing plant effluents generally contain high levels of soluble K^+ , making it readily available to plants. This effect was confirmed by the analyzed data, indicating that the effluent fertigation system contributed to an adequate supply of this macronutrient. However, the maintenance of these levels over time also indicates that the applied doses were compatible with crop demand and that the soil retention system was effective in preventing leaching losses.

Figures 2C and D show gradual increases in soil calcium and magnesium levels after effluent application, whereas the control group showed a slight reduction in these cations over time. Both nutrients play crucial roles in the cellular structure, membrane integrity, and enzyme function. Mombaca grass has a significant capacity for extracting Ca and Mg, as already pointed out by Euclides et al. (2021), especially in intensive management systems and under high growth rates.

The origin of base cations in the treated soil can be explained by the nature of the applied effluents.

Based on Zhao et al. (2021), combined industrial effluents, particularly those containing organic waste and neutralizing agents, often have a high load of exchangeable bases, which explains the increase in these elements. Furthermore, the balance between Ca^{2+} and Mg^{2+} is crucial to avoid antagonisms in root uptake, which appears to have been maintained in the treatments, given the similar behavior between the two nutrients.

Figures 3A–C show the evolution of potential cation exchange capacity (CECp), effective cation exchange capacity (CECe), and potential acidity ($H^+ + Al^{3+}$) over 1 year of cultivation with Mombaca grass.

A significant linear increase in the CECp and potential acidity was observed throughout the growing season. Although the CECe also showed an increasing trend, it did not show a statistically significant difference between the evaluated periods.

The increase in CECp over time can be attributed primarily to the increase in organic matter content and the input of basic cations from the applied effluents. Organic matter, especially in the form of organic waste of bovine origin, has a large number of active sites with cation adsorption and exchange capacities,

directly contributing to an increase in CEC (Costa et al., 2023). Furthermore, the input of complex organic compounds present in industrial effluents, such as polyphosphates and synthetic humic acids, from the treatment of metal surfaces in the paint industry can promote improvements in soil colloidal activity, as discussed by Zhang et al. (2022), resulting in a higher CECp.

Despite increasing over time, the CECe did not show statistically significant variations. This can be explained by the dynamic equilibrium between exchangeable cations in the solid phase and ions in solution, particularly considering the nutrient extraction regime imposed by the Mombaça grass crop. As previously discussed, the high nutritional requirements of this forage grass imply a high absorption rate of K^+ , Ca^{2+} , and Mg^{2+} , which may mitigate the positive effects of cation addition via effluent on the CECe (Euclides et al., 2021). The maintenance of stable CECe levels despite high absorption indicates a good balance between nutritional supply and demand in the system.

Potential acidity ($H^+ + Al^{3+}$) also showed a linear increase over time in the effluent treatments, indicating that, although there is a significant base input, there is also a simultaneous accumulation of protons and exchangeable aluminum in the sorption complex. This may be related to the mineralization of organic compounds present in the effluents, especially those from the paint industry, which can release short-chain organic acids during decomposition (Zhou et al., 2021). Furthermore, residues of dyes, pigments, and inorganic fillers that have not been fully degraded can interfere with acid–base balance of the soil, promoting, in some cases, an increase in latent acidity.

This behavior is consistent with the observed trend of a slight reduction in pH and base saturation in the final months of the experiment, revealing a possible slow and cumulative acidification of the soil, possibly because of the oxidation of organic compounds and the release of H^+ ions during the nitrification and degradation processes of organic matter. Based on Lima et al. (2023), the prolonged use of organic and industrial effluents can induce contrasting effects. Although promoting nutrient supply and increasing CEC, it can also acidify the soil in the long term, requiring corrective management.

In the agronomic context, an increase in CECp is a positive indicator of improved soil nutrient retention

capacity, which benefits intensive systems, such as forage production. However, the increase in potential acidity associated with a stable CECe indicates the need for constant monitoring and the possible application of acidity correctives (such as dolomitic limestone) to prevent aluminum saturation and reduce nutrient availability, especially phosphorus, whose adsorption intensifies in acidic environments.

4 Conclusion

Fertigation with combined effluent significantly increased the productivity of Mombaça grass, reaching an average of 18.28 t/ha of dry matter, indicating that the combination of the effluent did not result in phytotoxicity to the plant. Greater extraction of nitrogen, calcium, and magnesium was observed over 12 months, reflecting the nutritional efficiency of the treatments.

The agricultural reuse of effluents from the paint industry and cattle slaughter promoted an increase and stabilization of phosphorus, K, Ca, and Mg levels in the soil, even with intense extraction by Mombaça grass.

There was an increase in CECp, indicating an improvement in the nutrient retention capacity. However, an increase in potential acidity and slight reductions in pH and base saturation were also observed over time.

These results reveal the potential of effluent reuse as a sustainable alternative but reinforce the need for continuous monitoring and managing soil acidity.

The nutritional complementarity between effluents from cattle slaughter and the paint industry explains the observed agronomic efficiency. These results show the potential of this mixture as a sustainable source of nutrients for fertigation.

Future studies should evaluate the long-term effects on different soils and crops, in addition to investigating the behavior of potential contaminants, to ensure agronomic and environmental safety, primarily related to the accumulation of toxic metals in the soil and the transport of contaminants to water bodies.

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Code Availability Not applicable.

Declarations

Ethics Approval Not applicable.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing interests The authors declare no competing interests.

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